
Assignment 4.4

Batch-05

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1. Sentiment Classification for Customer Reviews

a) Customer Reviews with Sentiment Labels

Review	Sentiment
The product quality is excellent and delivery was fast.	Positive
I am very disappointed with the customer service.	Negative
The item is okay, nothing special.	Neutral
Loved the design and performance of the product.	Positive
The product stopped working after two days.	Negative
Packaging was fine, but delivery was delayed.	Neutral

b) Zero-shot Prompt

Classify the sentiment of the following review as Positive, Negative, or Neutral:

"The product quality is excellent and delivery was fast."

c) One-shot Prompt

Example:

Review: "The phone battery drains very fast."

Sentiment: Negative

Review: "The product quality is excellent and delivery was fast."

Sentiment: Positive

d) Few-shot Prompt

Examples:

Review: "Amazing build quality and great performance."

Sentiment: Positive

Review: "Worst experience, totally waste of money."

Sentiment: Negative

Review: "Product is average and acceptable."

Sentiment: Neutral

Review: "The product quality is excellent and delivery was fast."

Sentiment: Positive

e) Comparison & Accuracy Discussion

- **Zero-shot:** Works well but may misclassify neutral statements.
 - **One-shot:** Better accuracy due to reference example.
 - **Few-shot:** Most accurate and consistent due to multiple contextual examples
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2. Email Priority Classification

a) Sample Emails with Priority Labels

Email Content	Priority
Server is down, immediate fix required.	High

Email Content	Priority
Request for meeting reschedule next week.	Medium
Newsletter subscription confirmation.	Low
Client complaint regarding payment failure.	High
Monthly project progress update.	Medium
Holiday greeting email.	Low

b) Zero-shot Prompt

Classify the priority of this email as High, Medium, or Low:

"Server is down, immediate fix required."

c) One-shot Prompt

Example:

Email: "Urgent: Application crash affecting users"

Priority: High

Now classify:

Email: "Server is down, immediate fix required."

Priority: High

d) Few-shot Prompt

Examples:

Email: "Urgent: Payment gateway not working"

Priority: High

Email: "Please review the attached project report"

Priority: Medium

Email: "Company annual newsletter"

Priority: Low

Email: "Server is down, immediate fix required."

Priority: High

e) Evaluation

- **Zero-shot:** Quick but sometimes inconsistent.
- **One-shot:** Improved reliability.
- **Few-shot:** Most reliable due to clear priority patterns.

Few-shot prompting performs best.

3. Student Query Routing System

a) Sample Queries & Departments

Query	Department
What is the admission cutoff for CSE?	Admissions
When will semester results be announced?	Exams
What are the elective subjects this semester?	Academics
Companies visiting for campus placements?	Placements
How to apply for revaluation?	Exams
Internship credit rules?	Academics

b) Zero-shot Prompt

Route the following query to Admissions, Exams, Academics, or Placements:

"When will semester results be announced?"

c) One-shot Prompt

Example:

Query: "What is the admission cutoff?"

Department: Admissions

Now classify:

Query: "When will semester results be announced?"

Department:

d) Few-shot Prompt

Examples:

Query: "Placement training schedule?"

Department: Placements

Query: "How to register for electives?"

Department: Academics

Query: "Result rechecking procedure?"

Department: Exams

Query: "When will semester results be announced?"

Department: Exams

e) Analysis

- Contextual examples significantly improve intent recognition.

- Few-shot prompting reduces ambiguity.

Few-shot gives highest routing accuracy.

4. Chatbot Question Type Detection

a) Queries & Types

Query	Type
What is the refund policy?	Informational
I want to cancel my order.	Transactional
The app keeps crashing.	Complaint
Great experience using the app!	Feedback
How can I reset my password?	Informational
Delivery was late and disappointing.	Complaint

b) Prompt Designs

- **Zero-shot:** Direct classification without examples
 - **One-shot:** One labeled example provided
 - **Few-shot:** Multiple labeled examples covering all types
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c) Observations

- Zero-shot shows confusion between Complaint & Feedback.
- One-shot improves clarity.
- Few-shot handles ambiguity best.

Few-shot prompting ensures correct classification.

5. Emotion Detection in Text

a) Emotion Samples

Text	Emotion
I am feeling very happy today!	Happy
I feel lonely and sad.	Sad
This makes me so angry!	Angry
I am worried about my future.	Anxious
Today is just a normal day.	Neutral
I feel excited and joyful.	Happy

b) Zero-shot Prompt

Identify the emotion (Happy, Sad, Angry, Anxious, Neutral):

"I am worried about my future."

c) One-shot Prompt

Example:

Text: "I feel very low and depressed."

Emotion: Sad

Text: "I am worried about my future."

Emotion:

d) Few-shot Prompt

Examples:

Text: "I am extremely joyful today!"

Emotion: Happy

Text: "I am scared about tomorrow."

Emotion: Anxious

Text: "I feel irritated and mad."

Emotion: Angry

Text: "Nothing special happened today."

Emotion: Neutral

Text: "I am worried about my future."

Emotion: Anxious

e) Discussion

- Zero-shot struggles with mixed emotions.
- One-shot improves detection.
- Few-shot provides best emotional understanding.

Few-shot prompting handles ambiguity best.

Conclusion

Across all tasks:

- **Zero-shot** → Fast but less accurate
- **One-shot** → Moderate improvement
- **Few-shot** → Highest accuracy & consistency